AUTOMATED BACKUP AND RECOVERY SYSTEM

Services: EC2, EBS, Lambda, CloudWatch Events

Description: Automate EBS snapshot creation and deletion using Lambda triggered by CloudWatch Events. Implement snapshot retention policies.

Skills: Automation, backup strategies, Lambda scripting

1. Definition

The **Automated Backup and Recovery System** is a cloud-based solution designed to automatically create backups of storage volumes attached to servers and delete outdated backups based on a retention policy.

In this project, snapshots of Amazon Elastic Block Store volumes attached to Amazon EC2 instances are created automatically using a serverless function deployed on AWS Lambda. The function is triggered on a schedule using Amazon EventBridge (previously known as CloudWatch Events).

This automation eliminates manual backup processes and ensures data protection by maintaining periodic snapshots and removing old snapshots automatically.

2. Scope of the Project

The scope of this project includes:

- Automating the backup of EBS volumes attached to EC2 instances.
- Scheduling backups at regular intervals using EventBridge.
- Implementing snapshot retention policies.
- Reducing operational effort by eliminating manual backups.
- Ensuring cost optimization by automatically deleting outdated snapshots.
- Implementing tag-based filtering so that only selected volumes are backed up.

The project is suitable for environments where regular backups are required for disaster recovery and data protection.

3. Methodologies

The project follows an **event-driven automation methodology**.

Key Principles

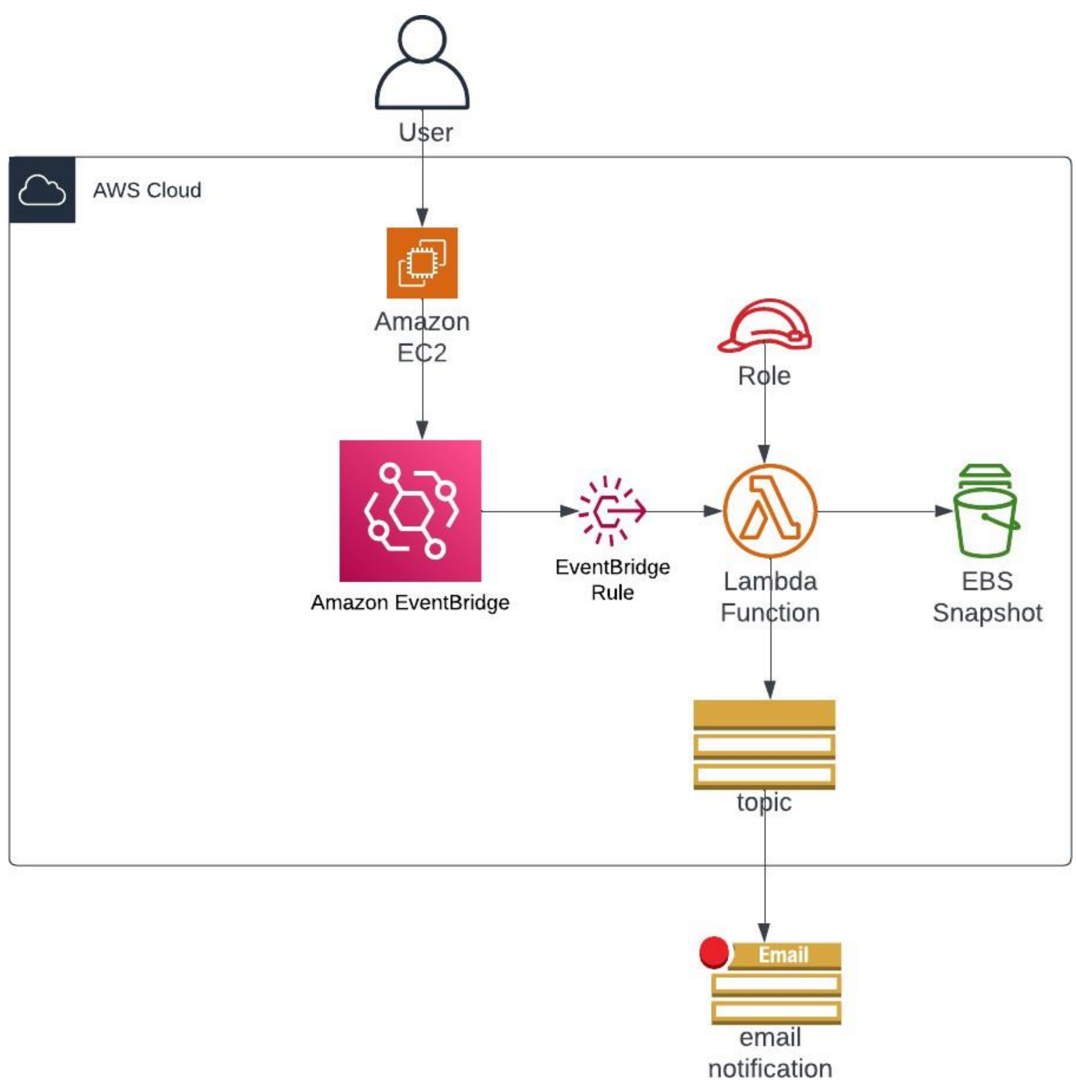
1. **Serverless Automation**
    - No servers required for running automation scripts.

- Lambda executes code only when triggered.

2. **Scheduled Event Trigger**
   - EventBridge scheduler triggers Lambda based on defined intervals.
3. **Tag-Based Resource Management**
   - Only volumes tagged with Backup=True are included in backups.
4. **Retention Policy**
   - Snapshots older than a defined period are automatically deleted.

This methodology ensures scalable, automated, and efficient backup management.

5. Architecture Diagram / Flow



Architecture Flow

EventBridge Scheduler

↓ (Scheduled Trigger)

AWS Lambda Function

↓ Uses IAM Role Permissions

Amazon EC2

↓

Amazon EBS Volumes

↓

EBS Snapshots Created

↓

Retention Policy
(Delete Old Snapshots)

Process Flow

1. EventBridge triggers Lambda based on schedule.
2. Lambda scans for EBS volumes with tag Backup=True.
3. Lambda creates snapshots of those volumes.
4. Lambda checks snapshot age.
5. Snapshots older than the retention period are deleted automatically.

5. Explanation of Services Used

1. Amazon EC2

Purpose:

- Provides virtual servers in the cloud.
- EC2 instances host applications and use EBS volumes for storage.
- In this project, EC2 instances have EBS volumes attached whose data must be backed up.

2. Amazon Elastic Block Store

Purpose:

- Provides persistent block storage for EC2 instances.

- Stores operating system, application files, and data.
- Snapshots are taken from EBS volumes to create backups.

3. AWS Lambda

Purpose:

- Executes automation scripts without managing servers.
- Runs Python code that:
    - Finds EBS volumes with specific tags
    - Creates snapshots
    - Deletes old snapshots based on retention rules.

4. Amazon EventBridge

Purpose:

- Schedules automated triggers for Lambda.
- Replaces CloudWatch Events.
- Ensures snapshots are created periodically (e.g., every day or every few minutes for testing).

5. IAM Role (AWS Identity and Access Management)

Purpose:

- Provides secure permissions to Lambda.
- Allows Lambda to interact with EC2 and EBS services.
- Permissions include:
    - Create snapshot
    - Delete snapshot
    - Describe volumes
    - Describe snapshots

6. Step-by-Step Procedure

Step 1: Launch EC2 Instance

1. Open AWS Console.
2. Navigate to EC2 service.
3. Click **Launch Instance**.
4. Choose Amazon Linux AMI.
5. Select instance type (t3.micro).
6. Configure network settings.
7. Configure storage (EBS volume attached automatically).
8. Launch the instance.

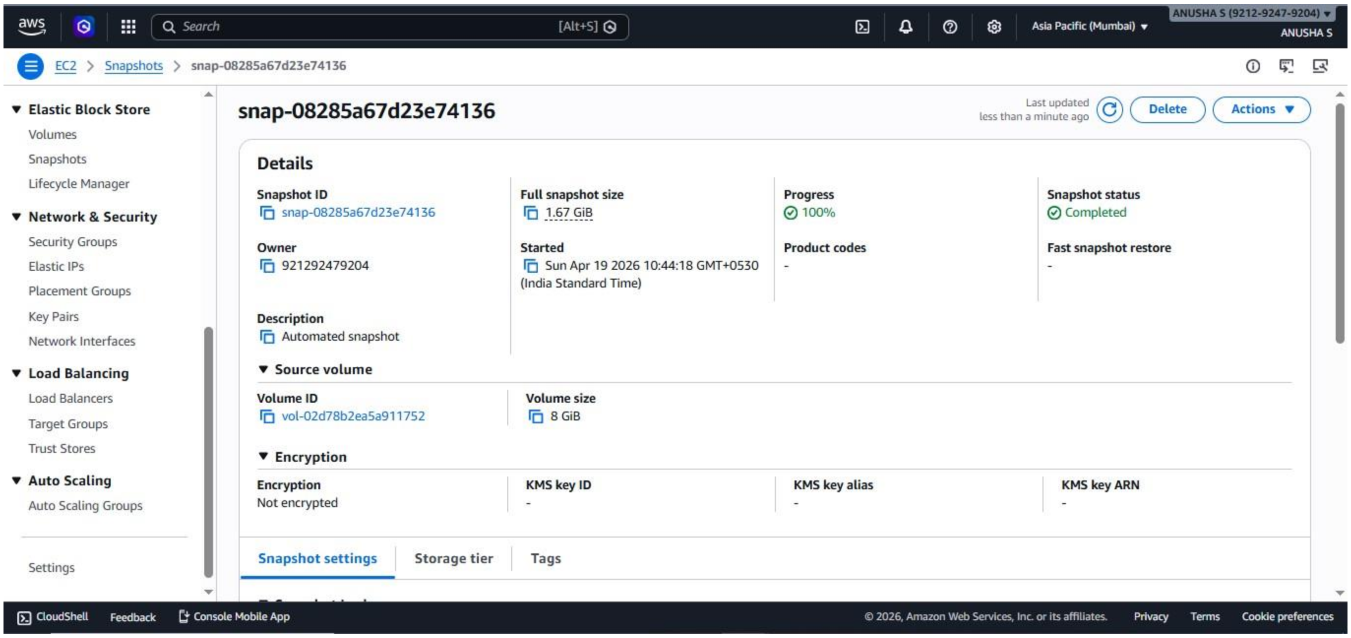
Result:
A server with an attached EBS volume is created.

Step 2: Tag the EBS Volume

1. Navigate to EC2 → Volumes.
2. Select the volume attached to the EC2 instance.
3. Click **Manage Tags**.
4. Add tag:

Key: Backup
Value: True

Purpose:
Ensures that only tagged volumes are included in the backup automation.



Step 3: Create IAM Role for Lambda

1. Navigate to IAM → Roles.
2. Click **Create Role**.
3. Select **Lambda** as trusted entity.
4. Attach policies that allow:
   - EC2 snapshot creation
   - Snapshot deletion
5. Name the role.

Purpose:
Allows Lambda to interact with EC2 and EBS securely.

Step 4: Create Lambda Function

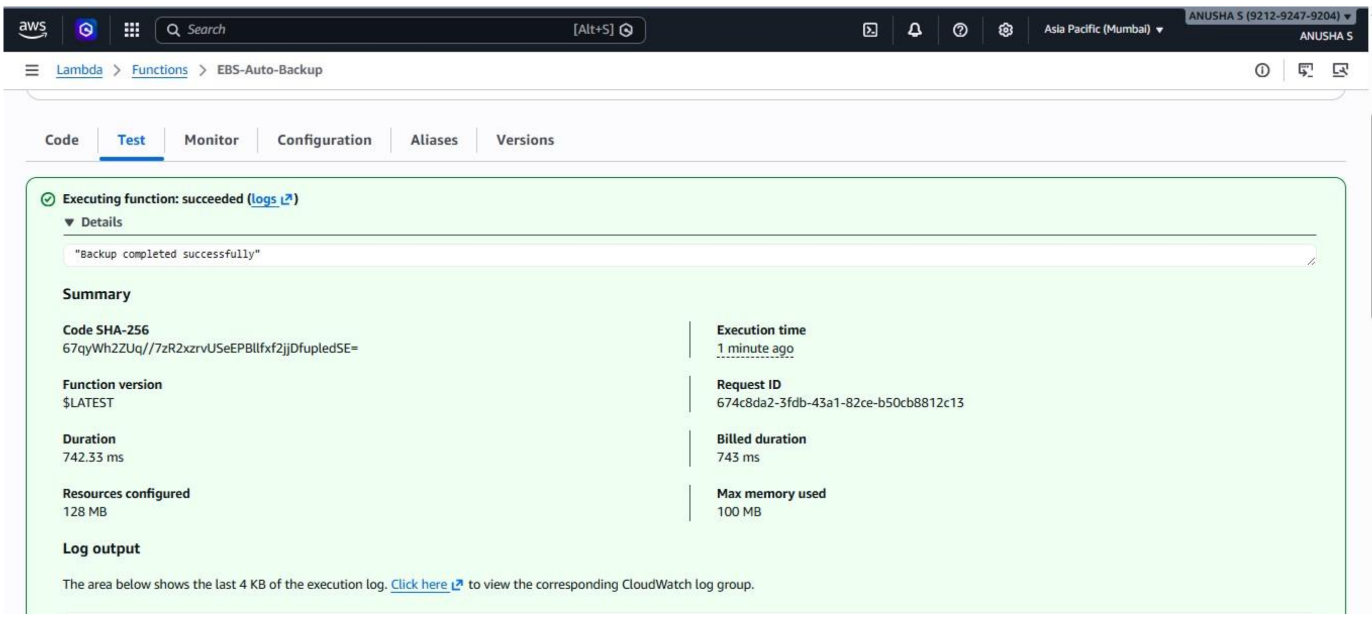
1. Navigate to Lambda service.
2. Click **Create Function**.
3. Select **Author from scratch**.

4. Choose Python runtime.
5. Attach IAM role created earlier.

Add Python code that:

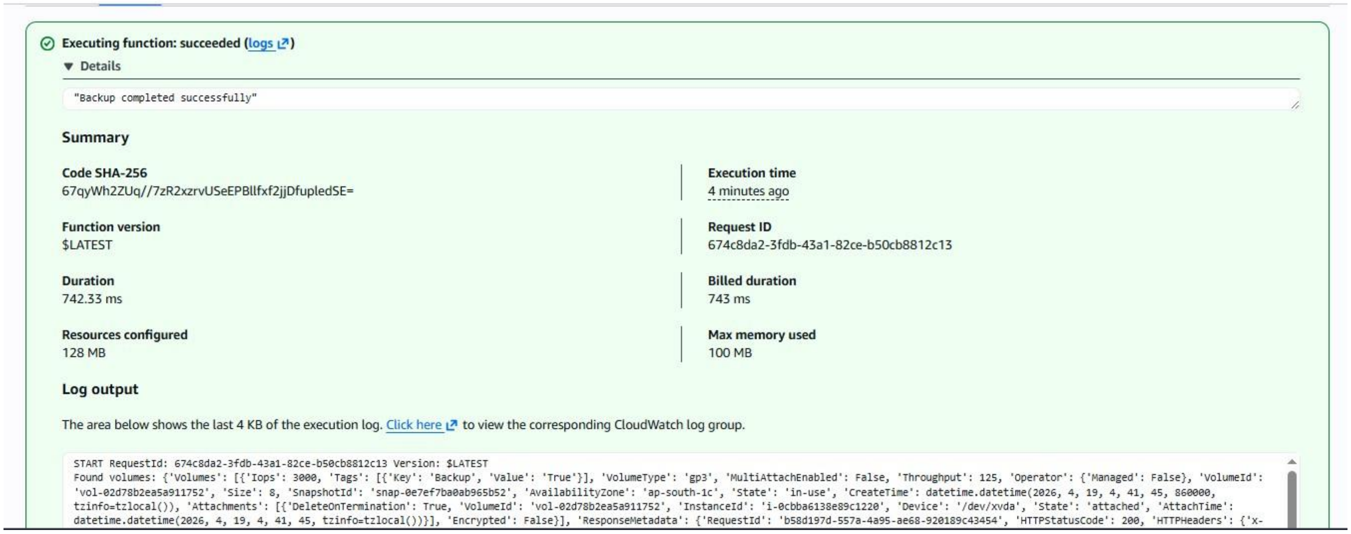
- Finds tagged volumes
- Creates snapshots
- Deletes snapshots older than retention period.

Deploy the function.



Step 5: Test Lambda Function

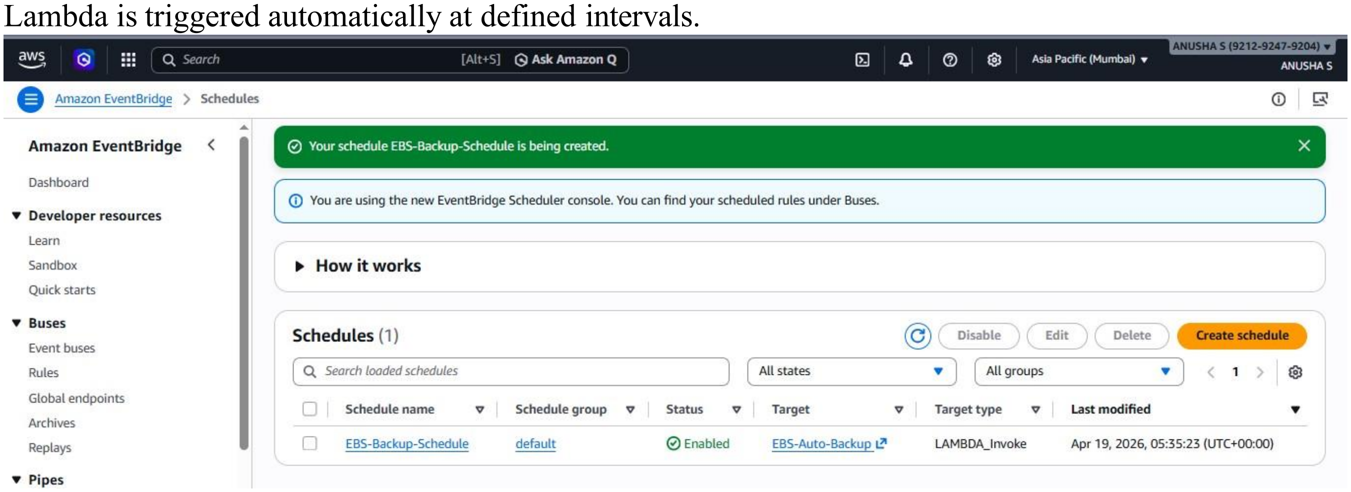
1. Click **Test** in Lambda console.
2. Execute the function.
3. Navigate to EC2 → Snapshots.
4. Verify that a new snapshot has been created.



Step 6: Configure EventBridge Scheduler

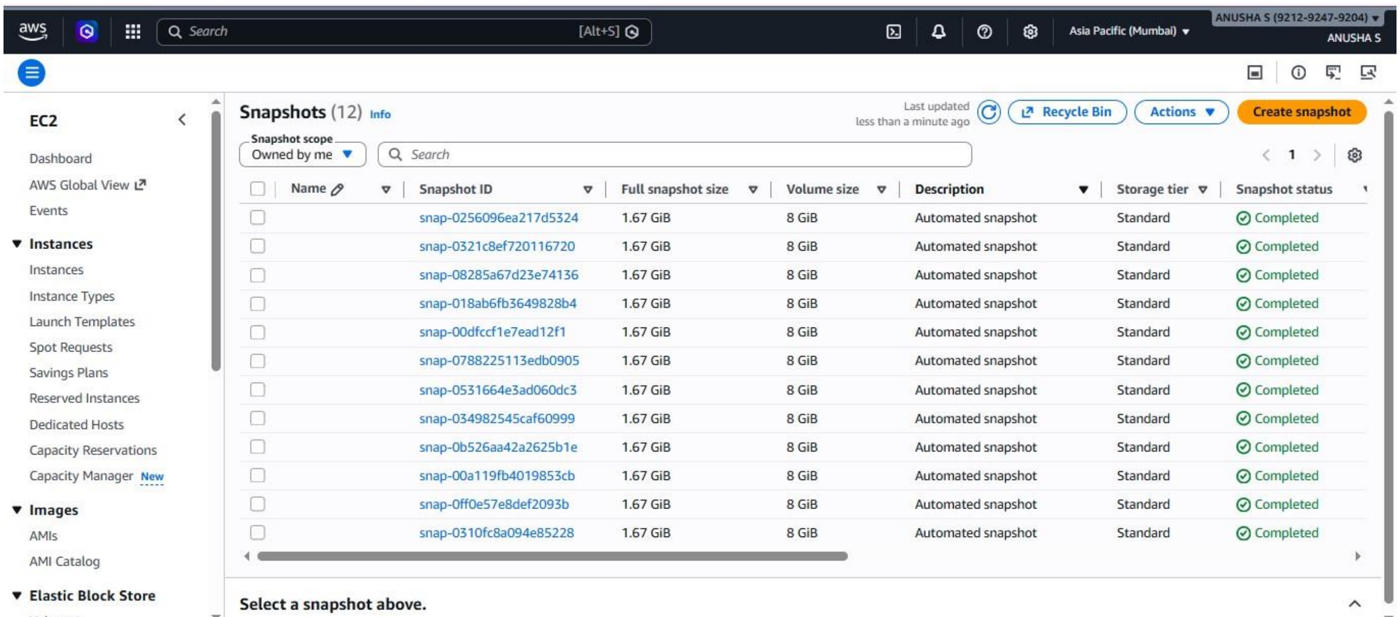
1. Navigate to EventBridge.
2. Click **Schedules**.
3. Create a new schedule.
4. Choose rate-based schedule (e.g., every 5 minutes for testing).
5. Select Lambda function as target.
6. Enable the schedule.

Result:
Lambda is triggered automatically at defined intervals.



Step 7: Verify Retention Policy

1. Lambda checks snapshot timestamps.
2. Snapshots older than defined period are deleted.
3. Logs confirm deletion events.



7. Troubleshooting

Issue 1: Lambda Cannot Create Snapshots

Method:
Check Lambda execution logs.

Solution:
Verify IAM role permissions include EC2 snapshot operations.

Issue 2: Snapshots Not Created

Method:
Check EventBridge schedule status.

Solution:
Ensure schedule is enabled and targeting correct Lambda function.

Issue 3: Snapshots Not Deleted

Method:
Check Lambda logs for deletion messages.

Solution:
Verify snapshot age and ensure snapshots contain the required tag.

Issue 4: Too Many Snapshots Created

Method:
Check schedule frequency.

Solution:
Change schedule interval from minutes to daily execution.

8. Conclusion

The **Automated Backup and Recovery System** successfully automates the backup process of EBS volumes using AWS serverless architecture.

By integrating EC2, EBS, Lambda, and EventBridge, the system ensures:

- Automated backup creation
- Controlled backup retention
- Reduced manual operational effort
- Cost optimization through automated cleanup

The implementation of tag-based filtering further improves production readiness by ensuring that only designated volumes are included in the backup process.

This project demonstrates practical knowledge of AWS automation, cloud infrastructure management, and serverless architecture for real-world backup solutions.